Erdős #400: the ledger, a short proof of the sharp upper bound, and the constant

Samuel Lavery

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Abstract

For $k \geq 2$ let $g_k(n) = \max\{(a_1 + \cdots + a_k) - n : \; a_1! \cdots a_k! \mid n!\}$. We record an exact reformulation of the divisibility as a per-rail digit-sum ledger, and from it give a five-line unconditional proof of $g_k(n) \leq (k-1)\log_2 n + \log_2 \log n + O_k(1)$, matching the upper bound of Li [1] — the bound comes from the prime 2 alone, because the ledger charges $p-1$ per carry and 2 is the only rail whose cell is trivial. We then give an exact polylogarithmic algorithm for the distribution of the carry count on each rail, validated against exhaustive computation, and use it to measure the constant of Erdős and Graham out to $n = 10^{31}$. The measurement says the upper bound is *sharp*:

$$\sum_{n \leq x} g_k(n) \sim c_k \, x \log x \quad \text{with} \quad c_k = \frac{k-1}{\log 2} = 1.44270\ldots(k-1),$$

against Li's proven bracket $c_k \in (k-1)[\,3/\log 12,\ 1/\log 2\,] = (k-1)[1.2073, 1.4427]$. The constant is **measured, not proved**; the upper bound and the ledger are proved.

1 The ledger

Write $s_p(m)$ for the sum of base-p digits of m. Let c_j denote the carry out of position j in the base-p addition $a_1 + \cdots + a_k$, and set

$$\kappa_p(a_1, \ldots, a_k) \;:=\; \sum_{j \geq 0} c_j,$$

the *total carry weight*. Throughout $A = a_1 + \cdots + a_k$ and $G = A - n$.

Remark 1 (weight, not count — a correction)**.** *For $k = 2$ every $c_j \in \{0, 1\}$ and κ_p is the* number *of carrying positions, which is Kummer's theorem in its usual form. For $k \geq 3$ a carry may exceed 1 (up to $k-1$, by Lemma 3), and the identity below holds for the total weight $\sum_j c_j$,* not *for the count of nonzero carries. An earlier draft of this note said "number of carries" throughout; that is correct only at $k = 2$. Caught in formalisation.*

Theorem 2 (ledger identity)**.** *For non-negative integers $a_1, \ldots, a_k$ and n,*

$$a_1! \cdots a_k! \;\Big|\; n! \qquad \Longleftrightarrow \qquad A - n \;\leq\; \sum_{i=1}^{k} s_p(a_i) - s_p(n) \quad \textit{for every prime } p.$$

Equivalently, since $\sum_i s_p(a_i) - s_p(A) = (p-1)\kappa_p$,

$$g_k(n) = \max_a \min_p \Big[(p-1)\,\kappa_p(a) \;+\; s_p(n+G) - s_p(n)\Big].$$

Proof. By Legendre, $v_p(m!) = \frac{m-s_p(m)}{p-1}$, so $v_p(n!) - \sum_i v_p(a_i!) = \frac{(n-s_p(n))-\sum_i(a_i-s_p(a_i))}{p-1} = \frac{\sum_i s_p(a_i)-s_p(n)-(A-n)}{p-1}$. Divisibility is exactly non-negativity of this for all p. The second form is Kummer's theorem $\sum_i s_p(a_i) - s_p(A) = (p-1)\kappa_p$. $\square$

Verified: 0 mismatches against brute-force factorial divisibility for all $n < 40$, $a_1 \le n+8$, $G \le 4$, with p ranging to $2n+8$. (Truncating the rail set is unsafe: with $p < 60$ only, g_2 is already wrong for 10 values of $n \le 400$.)

Two features of the ledger drive everything. First, *a carry on rail* p *pays* $p-1$, so the smallest rails bind: in computation $p = 2$ binds 73% of the time, $p = 3$ a further 20%. Second, every rail with $p > \max_i a_i$ contributes exactly G, so large rails are *always exactly tight* and the binding structure lives entirely on small rails.

2 The upper bound

Lemma 3 (universal carry cap)**.** *In the base-*p *addition of* k *non-negative integers, the carry out of every position is at most* $k-1$*. Consequently* $\kappa_p(a_1,\dots,a_k) \le (k-1)\big(\lfloor \log_p A\rfloor + 1\big)$.

Proof. Let c_j be the carry out of position j, $c_{-1} = 0$. Then $c_j = \big\lfloor(\sum_i a_{i,j} + c_{j-1})/p\big\rfloor \le \big\lfloor(k(p-1) + c_{j-1})/p\big\rfloor$. If $c_{j-1} \le k-1$ then the right side is at most $\lfloor(kp-k+k-1)/p\rfloor = \lfloor(kp-1)/p\rfloor = k-1$; induction gives $c_j \le k-1$ for all j. Positions beyond $\lfloor\log_p A\rfloor$ contribute no carry. $\square$

Verified: 40 000 random $(k, p, a_1, \dots, a_k)$ with $k \le 6$, $p \le 11$, $a_i < 10^8$: no violation of either statement.

Theorem 4 (sharp upper bound)**.** *For every* $k \ge 2$ *and all large* n,

$$g_k(n) \;\le\; (k-1)\log_2 n + \log_2\log_2 n + O_k(1), \qquad \textit{hence} \qquad c_k \le \frac{k-1}{\log 2}.$$

Proof. Take $p = 2$ in Theorem 2. Since $p-1 = 1$,

$$G \;\le\; \kappa_2 + s_2(A) - s_2(n).$$

By Lemma 3, $\kappa_2 \le (k-1)(\lfloor\log_2 A\rfloor + 1)$. By subadditivity of the binary digit sum, $s_2(A) = s_2(n+G) \le s_2(n) + s_2(G)$, so $s_2(A) - s_2(n) \le s_2(G) \le \log_2(G+1)$. Therefore

$$G \;\le\; (k-1)\log_2 A + \log_2(G+1) + O(k). \tag{1}$$

It remains to remove the G on the right, and (1) does this by itself. Write $C = (k-1)\log_2 A + O(k)$, so that $G \le C + \log_2(G+1)$. Since $G \le A$ we get the crude bound $G \le C + \log_2(A+1)$, hence $G = O_k(\log A)$; feeding that back into (1) gives $\log_2(G+1) \le \log_2\log_2 A + O_k(1)$ and therefore

$$G \;\le\; (k-1)\log_2 A + \log_2\log_2 A + O_k(1).$$

Finally $\log_2 A = \log_2 n + O(1/n)$ because $A = n + G$ with $G = O_k(\log n)$. $\square$

Remark 5 (no external input is needed)**.** *An earlier draft absorbed the* $\log_2(G+1)$ *term by quoting Erdős and Graham's bound* $g_k(n) \ll_k \log n$*. That citation is unnecessary: as above,* (1) *bootstraps against itself in one step, since* $G \le A$ *already gives* $G = O_k(\log A)$*. So Theorem 4 is self-contained, depending only on Legendre, Kummer and Lemma 3. Noticed in formalisation, and the formal proof takes this route.*

Remark 6. *The bound is forced by a single rail, and by the* only *rail whose cell* $p-1$ *equals* 1*: for general* p *the same argument gives* $G \le (p-1)(k-1)\log_p A + O(\log\log n)$*, and* $\min_p \frac{(p-1)}{\log p}$ *is attained at* $p = 2$ *with value* $1/\log 2$*. Deleting rail* 2 *from the constraint set raises the measured constant by a factor* 1.35*; restricting to* $p \equiv 1 \pmod 6$ *raises it by* 2.95*.*

Remark 7 (the cap is attained)**.** *The carry recursion* $c = \lfloor (k+c)/2 \rfloor$ *has fixed point* $c = k-1$*, realised when every binary digit of every* a_i *equals* 1*, i.e.* $a_i = 2^m - 1$*. So Lemma 3 is sharp on rail* 2*, and the upper bound of Theorem 4 is attainable* as a constraint*. Whether such splits are available for a given* n*, subject to the remaining rails, is the lower-bound problem; the best proved lower bound is Li's* $\left(\frac{3(k-1)}{\log 12} - \varepsilon\right)\log n$ *for density-one* n *[1].*

3 Monotonicity in the number of parts

The upper bound of §2 is complemented by an exact monotonicity, proved by appending a trivial part.

Theorem 8 (append a one)**.** *For all* $n \ge 1$ *and all* $k \ge 3$*,*

$$g_k(n) \;\ge\; g_{k-1}(n) + 1,$$

and consequently $g_k(n) \ge g_2(n) + k - 2 \ge k - 1$ *for every* $k \ge 2$*.*

Proof. Let $b_1, \ldots, b_{k-1}$ attain $g_{k-1}(n)$, so $b_1! \cdots b_{k-1}! \mid n!$ and $\sum_{i<k} b_i = n + g_{k-1}(n)$. Put $a_i = b_i$ for $i \le k-1$ and $a_k = 1$. Then $\prod_{i\le k} a_i! = \prod_{i<k} b_i!$ divides $n!$, so the k-tuple is admissible, while $\sum_{i\le k} a_i - n = g_{k-1}(n) + 1$. As $g_k(n)$ is a supremum over admissible k-tuples, the inequality follows. Iterating down to $k = 2$ gives $g_k(n) \ge g_2(n) + k - 2$; and $b_1 = n$, $b_2 = 1$ is admissible with $b_1 + b_2 - n = 1$, so $g_2(n) \ge 1$. □

Remark 9 (why the step is exactly tight)**.** *Read on the ledger of Theorem 2, appending* $a_k = 1$ *raises* $s_p(a_k)$ *by exactly* 1 *on* every *rail simultaneously, while* G *also rises by exactly* 1*. So the constraint* $G \le \min_p\left[\sum_i s_p(a_i) - s_p(n)\right]$ *is preserved with equality in the increment, rail by rail: no rail is left slack and none is violated. This is why* 1 *is the correct part to append and why the step cannot be improved by another choice of* a_k *— any* $a_k \ge 2$ *raises* G *by* a_k *but raises* $s_p(a_k)$ *by only* a_k *for* $p > a_k$ *and by strictly less for* $p \le a_k$*, so some small rail loses.*

Corollary 10. $c_k \ge c_2$ *for every* $k \ge 2$*, so with Theorem 4,* $c_2 \le c_k \le \frac{k-1}{\log 2}$*. In particular the* k*-dependence of Erdős and Graham's* $g_k(n) \ll_k \log n$ *cannot be made uniform in* k*.*

Remark 11 (scope of the append step)**.** *The move applies exactly when the quantity is a supremum over tuples whose admissibility is a divisibility between products of factorials, since then* $1! = 1$ *is a free unit. It therefore transfers to Erdős #391 (largest least part* *in a factorisation $n! = a_1 \cdots a_n$), giving monotonicity in the number of parts by the same argument. It applies only one-sidedly to #374 ($a_1 < \cdots < a_k = m$ with $a_1! \cdots a_k!$ a square): appending 1 preserves squareness because $w(1) = 0$ in the $\mathbb{F}_2$ valuation ledger, but strict increase permits it only at the bottom and only when $a_1 > 1$. It does not apply to #683, #685, #700, #727, #1093, #1094, #1095, where the quantity is a least n, an extreme prime factor, a divisor count or a gcd minimum: those load every rail onto a single carrier point, so there is no tuple coordinate to extend.*

4 An exact algorithm, and the constant

Proposition 12 (carry automaton)**.** *For fixed A and p, the exact distribution of κ_p over all splits $a_1 + a_2 = A$ is computed in $O(\log_p^2 A)$ arithmetic operations by dynamic programming on (position, carry, #carries): with A_j the j-th digit of A, a transition from carry-in c to carry-out c' at position j is weighted by the number of pairs $(x, y) \in [0, p-1]^2$ with $x + y = T := A_j + pc' - c$, namely $T + 1$ if $T \leq p - 1$ and $2p - 1 - T$ otherwise.*

Verified against brute force for all $A < 180$ and $p \in \{2, 3, 5, 7\}$.

Writing $R_p = \lceil (G - (s_p(A) - s_p(n)))/(p-1) \rceil$ for the carry demand on rail p and $\mathrm{surv}_p(G)$ for the DP probability that $\kappa_p \geq R_p$, the first-moment estimator is

$$\widehat{g}_k(n) = \max\Big\{G : \ (A+1) \prod_p \mathrm{surv}_p(G) \ \geq \ 1\Big\}.$$

Two facts license it. The cross-rail correlation factor between the true joint survival and the product is $\Gamma = 1.03 \pm 0.34$ over $n \leq 1200$ — the rails are effectively independent. And $\widehat{g}_2$ is *unbiased* against exhaustive g_2 on $n \in [300, 1600]$: mean bias -0.05, ratio 0.9943, agreement within ± 1 pointwise.

Measured. With $\widehat{g}_2$ evaluated on random n at each scale ($\log n$ from 10 to 70):

n	10^{12}	10^{15}	10^{18}	10^{22}	10^{26}	10^{30}
$\widehat{g}_2/\log n$	1.4022	1.4070	1.4136	1.4325	1.4352	1.4229
gap to $1/\log 2$	-0.041	-0.036	-0.029	-0.010	-0.008	-0.020

The gap to $1/\log 2 = 1.44270$ shrinks; the gap to $3/\log 12 = 1.20729$ stays near $+0.21$ throughout. Hence the prediction

$$\boxed{c_k = \frac{k-1}{\log 2}}$$

i.e. Li's upper bound is sharp and his density-one lower bound is not the truth.

Remark 13 (why earlier attempts missed it)**.** *Two errors, both instructive. Estimating c_2 by averaging per-n maxima drifts and does not converge; averaging the occupancy* first *and taking the quantile of the mean is stable. And over $n \leq 10^4$ the functions $\log_2 n$ and $\log_2 \log n$ are nearly collinear, so a fit on that range cannot separate 1.2073 from 1.4427; the carry automaton removes the enumeration cost and lengthens the lever from $\log n \in [5.6, 8.8]$ to $[10, 70]$, at which point the separation is unambiguous.*

Remark 14 (register)**.** ***Proved:*** *Theorem 2, Lemma 3, Theorem 4, Theorem 8, Corollary 10, Proposition 12. Theorem 4 recovers Li's upper bound by a different and shorter route; no priority is claimed for it. Theorem 8 is elementary and gives* $c_2 \le c_k \le (k-1)/\log 2$*.* ***Measured:*** $c_k = (k-1)/\log 2$*, with the three validations above.* ***Not proved:*** *the lower bound, hence #400 itself.* ***Falsifier, pre-registered:*** $\widehat{g}_2/\log n$ *settling below* 1.40 *at* $n \ge 10^{40}$*, or* $\widehat{g}_2$ *developing bias against exhaustive* g_2*.*

References

[1] E. Li, *Prime-power rarefaction and a density-one lower bound for Erdős problem 400*, arXiv:2606.23661v2 (2026).

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monogr. Enseign. Math. 28 (1980), p. 77.